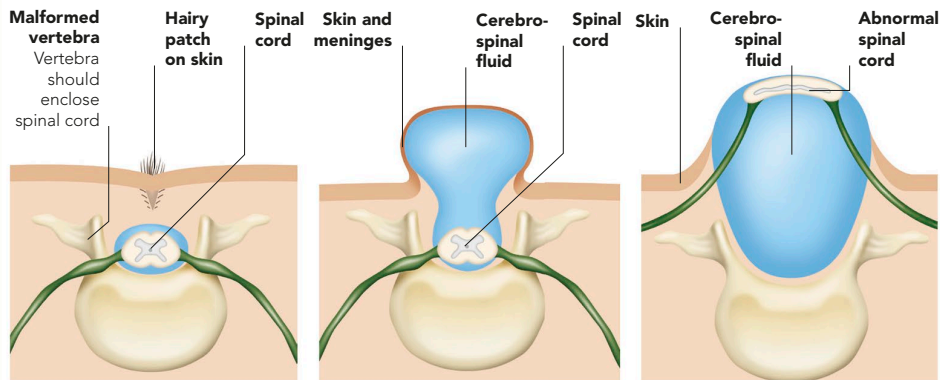


SPINA BIFIDA

There are three main forms of spina bifida: spina bifida occulta, meningocele, and myelomeningocele. Spina bifida occulta may require surgery to avoid serious neurological complications later in life. Meningoceles usually have a good prognosis after surgery. Myelomeningocele has effects that may include

paralysis or weakness in the legs, and lack of bladder and bowel control. Children with this form will have a permanent disability and may need extra support during their lives. Folic acid helps to prevent spina bifida, and women are advised to take supplements when planning to conceive and during the first 12 weeks of pregnancy.



SPINA BIFIDA OCCULTA

One or more vertebrae in the spine are malformed, but the cord is not damaged. On the skin, there may be dimpling, a tuft of hair, a fatty lump (lipoma), or a birthmark.

MENINGOCELE

The meninges protrude through a malformed vertebra as a visible fluid-filled sac called a meningocele. The spinal cord remains intact, and the defect can be repaired.

MYELOMENINGOCELE

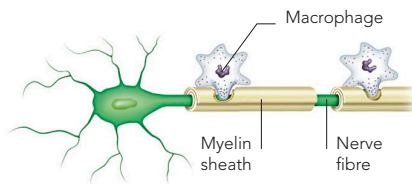
A part of the spinal cord, contained within a sac of fluid, protrudes through the skin. This is the most severe form of spina bifida, and will leave the child with some degree of disability.

MULTIPLE SCLEROSIS

Multiple sclerosis (MS) is due to immune system damage to the sheaths that insulate nerve fibres. It affects sensation, movement, body functions, and balance. In some people, symptoms may last for days or weeks, then clear up for months or years. In others, they gradually get worse. Drugs may help to lengthen remission periods and shorten attacks.

EARLY STAGE

At first, there are only small patches of damage. Macrophages, a type of scavenging cell, remove damaged areas of the myelin sheaths, exposing the fibres and impairing nerve conduction.



LATE STAGE

As MS progresses, the amount of damage to the sheaths increases, and affects conduction in more fibres. As the damage spreads, the symptoms become progressively worse.

